

PROBLEM SET 10

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 20)

1) **(10 pts)** Note that the angular momentum seems to generate a repulsive potential term in Eq. (12.3.13).

(a) **(5 pts)** Identify this repulsive term $V_{\text{rep}}(\rho)$

12.3.5

$$\left[-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{d\rho^2} + \frac{1}{\rho} \frac{d}{d\rho} - \frac{m^2}{\rho^2} \right) + V(\rho) \right] R_{Em}(\rho) = E R_{Em}(\rho)$$

↑ generated by angular momentum

$$V_{\text{ang.}} = \frac{\hbar^2 m^2}{2\mu \rho^2} \quad \text{"repulsive term"}$$

(b) **(5 pts)** Calculate the gradient of this repulsive term (identified as the centrifugal force)

$$V_{\text{ang.}} = \frac{\hbar^2 m^2}{2\mu \rho^2}$$

$$\vec{F} = -\nabla V_{\text{ang.}} = \frac{\hbar^2 m^2}{\mu \rho^3} \hat{\rho} \quad (\text{repulsive force})$$

$$\text{Since } L_z = m\hbar, \text{ we have } F = \frac{L_z^2}{\mu \rho^3}$$

2) **(10 pts)** Consider a particle of mass μ constrained to move on a circle of radius a .

(a) **(5 pts)** Show that $H = \frac{L_z^2}{2\mu a^2}$

Hint: note that the particle only has kinetic energy, hence you can start with $H = P^2/2\mu$, where $P \rightarrow$ momentum operator in polar coordinates.

$$12.3.6/ \quad H = \frac{P^2}{2\mu} = -\frac{\hbar^2}{2\mu} \vec{\nabla}^2 = -\frac{\hbar^2}{2\mu} \left(\vec{e}_\varphi \frac{1}{a} \frac{\partial}{\partial \varphi} \right)^2 = -\frac{\hbar^2}{2\mu a^2} \frac{\partial^2}{\partial \varphi^2}$$

Here we used $\vec{\nabla} = \vec{e}_r \frac{\partial}{\partial r} + \vec{e}_\varphi \frac{1}{a} \frac{\partial}{\partial \varphi}$ in polar coordinates.

But because of the constraint, φ is the only coordinate any wave functions would depend on, so $\vec{\nabla} = \vec{e}_\varphi \frac{1}{a} \frac{\partial}{\partial \varphi}$.

On the other hand $L_z^2 = -\hbar^2 \frac{\partial^2}{\partial \varphi^2}$. It is then clear

$$1) \quad H = \frac{L_z^2}{2\mu a^2}$$

- (b) **(5 pts)** Solve the eigenvalue problem of H to show that $E_m = \frac{\hbar^2 m^2}{2\mu a^2}$ and interpret the degeneracy (what is the degeneracy value?)

Its eigenvalue problem

$$\hat{H}|\Phi\rangle = E|\Phi\rangle$$

reads in position eigenbasis

$$\frac{1}{2\mu a^2} \left(-\hbar^2 \frac{\partial^2}{\partial \varphi^2} \right) \Phi(\varphi) = E \Phi(\varphi)$$

$$\Rightarrow \frac{\partial^2 \Phi(\varphi)}{\partial \varphi^2} = -\frac{2\mu a^2 E}{\hbar^2} \Phi(\varphi)$$

Solutions are $\Phi_m(\varphi) = e^{\pm im\varphi}$ ($m = 0, 1, 2, \dots$)

$$\text{with } m^2 = \frac{2\mu a^2 E_m}{\hbar^2} \Rightarrow \boxed{E_m = \frac{\hbar^2 m^2}{2\mu a^2}}$$

So the energy eigenvalues are $E_m = \frac{(L_z)_m^2}{2\mu a^2}$ with $(L_z)_m = m\hbar$

Each energy eigenvalue is two-fold degenerate:

$$E_m = E_{-m}$$

The two eigenfunctions corresponding to E_m describe clockwise and counterclockwise motion of the particle around the circle.